

Ananthu Rajendran Pillai

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Skills

Programming Languages: Python, C++, SQL, Matlab

Frameworks & Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, Hugging Face Transformers, LangChain, LangGraph

Tools & Technologies: Git, Linux, LaTeX, MySQL, Faiss, Ptc Creo, MS Excel

Experience

Research Intern, A*star – Singapore

June 2025 – Mar 2026

- Engineered **TS2 (Train-Sparsemax+, Test-Softmax)**, a novel **supervised finetuning framework** for LLMs that decouples training and inference to maximize output diversity without accuracy regression.
- Achieved a **+11.5% relative winrate** on AlpacaEval versus state of the art baselines and a **+25.1% increase in BLEU-based** diversity metrics.
- Maintained **zero shot utility across rigorous standard benchmarks (MMLU, ARC, HumanEval pass@k)** by mathematically resolving the traditional trade off between model winrate and response diversity.
- Investigated preference alignment learning dynamics and **squeezing effect within DPO and GRPO** frameworks.
- Co-authored conference paper: TS2: Training with Sparsemax+, Testing with Softmax for Accurate and Diverse LLM Fine-Tuning (Accepted at ICLR 2026).**

Research Intern, Energy Research Institute @ Ntu – Singapore

Mar 2024 – Dec 2024

- Engineered an LSTM-driven Model Predictive Control (MPC) system for building HVAC optimization, achieving a validated **25–40% reduction in energy consumption**.
- Integrated machine learning based **MPC algorithm** into a BACnet-based BMS in collaboration with Ngee Ann Polytechnic, leveraging MySQL for scalable deployment.
- Set up the deployment environment using **Linux and MySQL** to handle data flow and support real time model inference.

Electromechanical Design Engineer, Schneider Electric – Coimbatore, India

July 2022 – Aug 2023

- Led R&D optimization for the TFTQ low voltage switchboard line using PTC Creo Parametric, delivering a **closed door operation (CDO+)** feature that increased orders by **20% and ensured IEC 61439 compliance**.
- Integrated temperature & humidity sensors for real time thermal monitoring, improving reliability and enabling preventive maintenance workflows.

Education

Nanyang Technological University, MSc in Computer Control & Automation – Singapore
CGPA: 4.56/5.0

Jan 2024 – Jan 2026

- Relevant Coursework: **Genetic Algorithms & Machine Learning, AI & Data Mining, Neural Networks & Deep Learning, Machine Vision, Natural Language Processing, Robotics & Intelligent Sensors.**

College of Engineering, Trivandrum, BTech in Electrical & Electronics Engineering – India
CGPA: 9.71/10.0

Aug 2018 – July 2022

- Relevant Coursework: **Linear Algebra, Probability & Statistics, Calculus, Signals & Systems, Analog & Digital Electronics.**

Projects

ML-Based Semantic Description for Smart Buildings

Python, Ultralytics YOLOv8, Mistral-7B, Gemini 2.0 Flash, FAISS, RAG, JSON, Brick Schema

- Built a multi stage pipeline to convert service building drawings PDFs/images into Brick Schema.

- Detected components using **YOLOv8**, built relationships among them via heuristics & LLM prompting to produce a JSON graph.
- Employed an **RAG enhanced LLM** (with Brick ontology docs) to generate the final Brick Schema file.
- Evaluated against manually curated schemas: **Avg F1 (classes): 93.85%, Avg F1 (edges): 80.69%**.

Resume Job Description Matcher

Python, Streamlit, Hugging Face Transformers, PyTorch

- Deployed a Streamlit app which classifies resumes into 24 job categories, matches resumes to job descriptions as No Fit / Potential Fit / Good Fit, parses skills via spaCy NER to highlight gaps & recommend keywords.
- Fine tuned BERT (Hugging Face) on matching and NER tasks, achieving robust precision/recall.

Human Action Recognition in Low-Light Conditions

Python, NumPy, Pandas, Scikit-learn, PyTorch (ViT)

- Designed a video classification pipeline for 2–3 s clips under poor illumination: frame sampling, feature extraction, normalization, classification (Gaussian Naive Bayes, Logistic Regression).
- Prototyped Vision Transformer model in PyTorch to push beyond traditional ML baselines.

Certifications and Extracurriculars

- Programming for Everybody in Python, Coursera.
- Supervised Machine Learning: Regression and Classification, Coursera.
- NTU GSA Administration Subcommittee Member (organized volunteering & team-building).
- National Service Scheme Volunteer (240 hours community service).